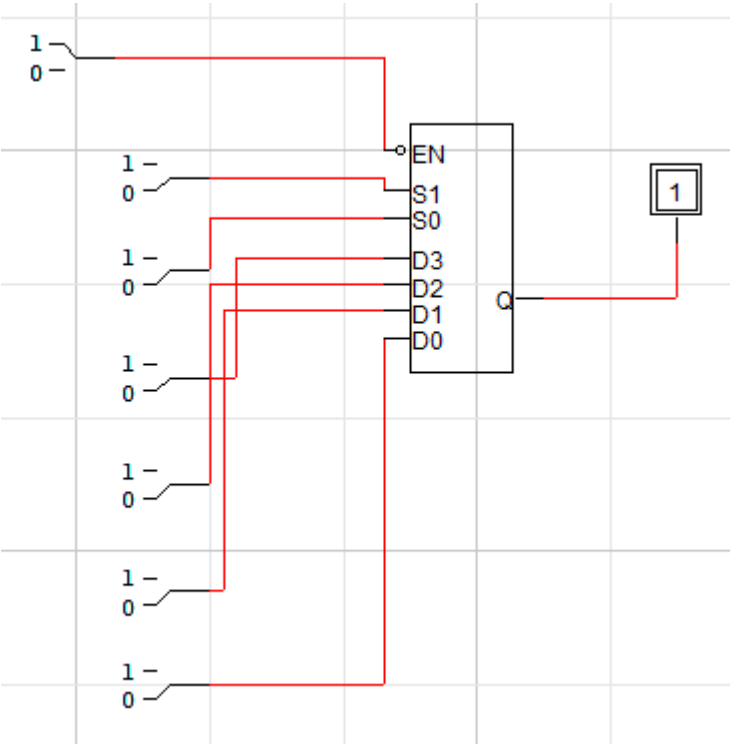
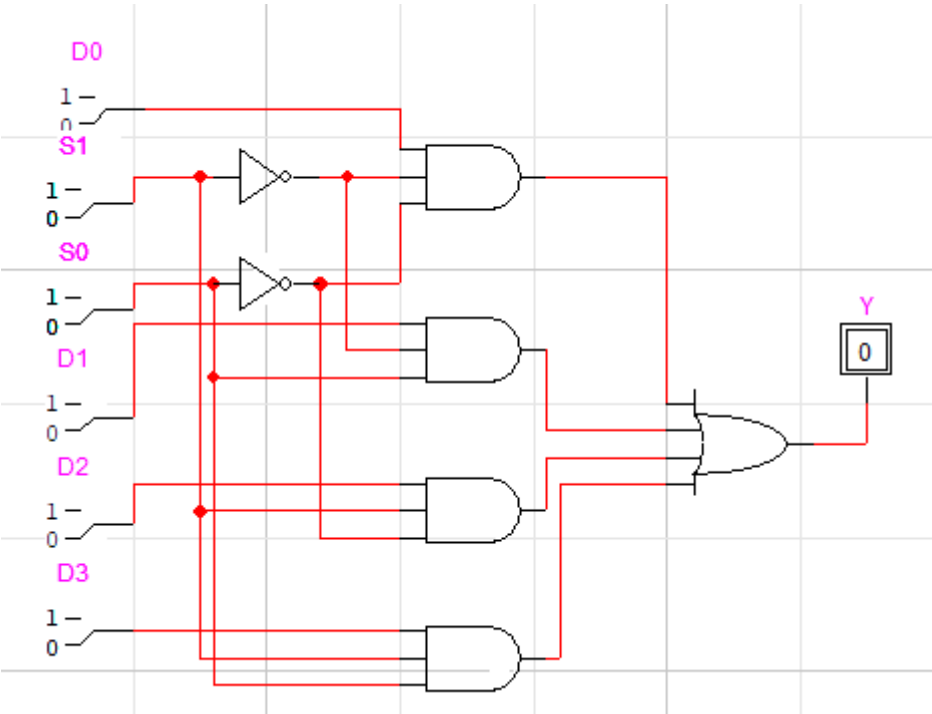


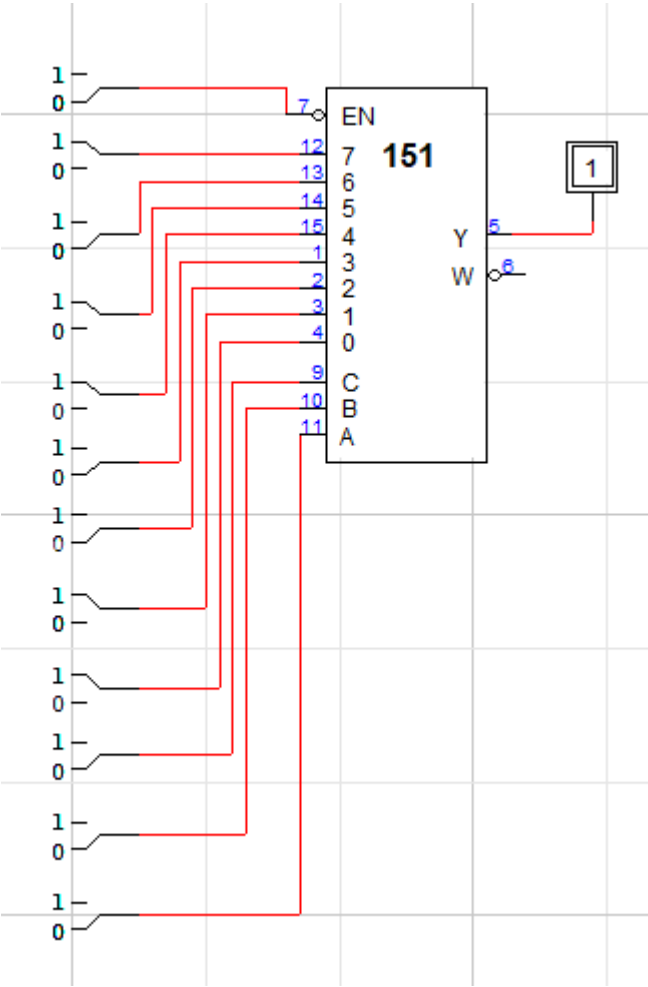
Task 1:



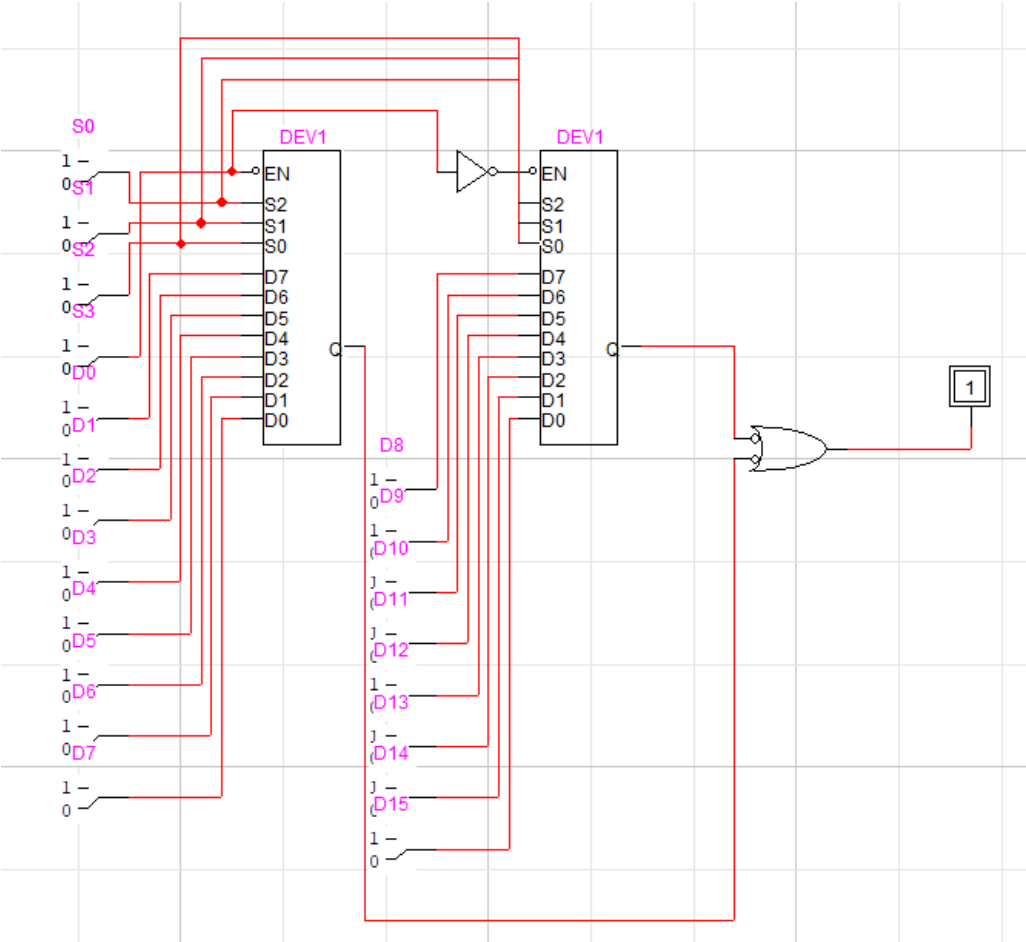
Task 2:



Task 3:



Task 4:



Task 5:

A seven segment display multiplexer is used to control multiple displays using fewer pins.

It works by rapidly switching (multiplexing) between displays so fast that all appear ON simultaneously.

A multiplexer (MUX) selects one set of BCD input at a time. The selected input is sent to a BCD to a seven segment decoder, which converts it into segment signals.

A control signal (clock) continuously switches between digits (MSD and LSD). At the same time, a decoder enables only one display at a time. Because switching is very fast, human eyes see both digits continuously lit. This reduces hardware complexity and power consumption. It is commonly used in digital clocks, calculator, and counters.

Task 6:

This system transmits data with error checking using parity bits. At the sender side, multiple data inputs are selected using a multiplexer (MUX). An even parity generator adds a parity bit to the data before transmission. The data and parity bits are sent together through the transmission line. At the receiver side, a demultiplexer (DEMUX) separates the received data. The data is stored and also sent to a parity checker. The parity checker recalculates parity and compares it with the received parity bit. If both match, the data is correct otherwise an error is detected. An error gate outputs "1" to indicate an error. This method ensures reliable data communication.

Task 7:

The tablet bottling system begins with the keypad input, which is encoded into BCD and stored in Register A. The value represents the tablets per bottle and is decoded for display. Meanwhile, a sensor and valve feed pulses into an 8-bit counter, which tracks the actual tablets being filled. The counter's binary output is sent to the adder block, where it is added to the 16-bit running total in register B. The adder is implemented using cascaded 74HC283 4-bit adders, with proper carry chaining to handle the full 16-bit sum. The updated total is stored back in register B, ensuring continuous accumulation. A comparator checks register A against register B to control bottle filling. Finally, register B's binary total is converted to BCD, decoded and displayed as the total tablets bottled. This integration of keypad input, counting, addition, and display ensures accurate monitoring and automation of the bottling process.

Task 8:**a):**

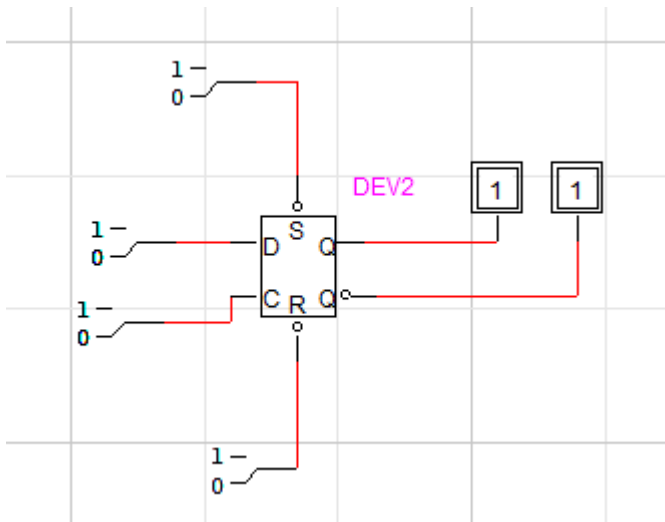
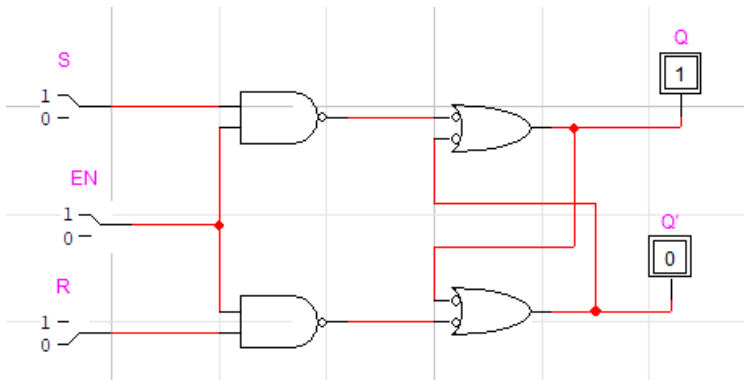
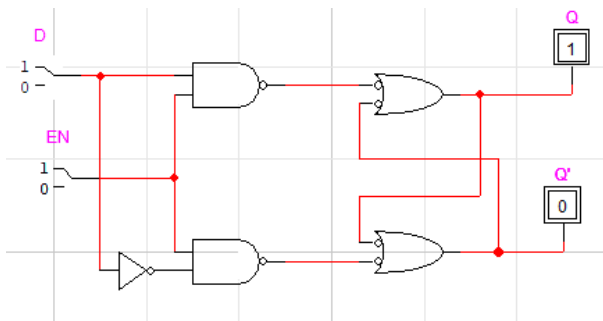
When S goes LOW, the latch sets and output $Q=1$. This state remains even after S returns High. If the latch was already set ($Q=1$), no change occurs because the output is stable.

b):

When R goes LOW, the latch resets and output becomes $Q=0$. This state remains even after R turns HIGH. If the latch was already reset ($Q=0$), no change occurs because the output is stable.

C):

When both S and R are HIGH, the latch holds its previous state. If it was SET, it stays SET, if it was RESET, it stays RESET. When both S and R go LOW at the same time, the outputs conflict. When inputs return HIGH together, the final state is uncertain and undefined.

Task 9:**Task 10:****Task 11:**

Task 12: